Customers Like You: Recommendation System Summary

Recommendation

Truncated SVD was selected because the customer product matrix is highly sparse, and SVD can compress this type of data efficiently.

Results

The original customer-product matrix would require about 189.83 MB if stored densely. Using 100 SVD components reduces this to about 7.65 MB, a 95.97% reduction in storage size.

The recommendation system achieved a Hit Rate@10 of 20.26%, compared with 17.25% for recommending the 10 best-selling products. Generating top-10 recommendations for 1,930 customers took 1.49 seconds, showing that the personalized model outperformed the baseline while remaining fast enough for a nightly recommendation process.

PCA produced similar recommendation accuracy, but SVD was preferred because it is better suited to the sparse purchase data.

Compression Loss

Compression reduces thousands of product dimensions to 100 components. This makes the data easier to store and process, but some information from the original purchase matrix is lost. At 100 components, the SVD model had a relative reconstruction error of 0.4961.

The goal is not to reproduce every purchase exactly, but to preserve enough of the underlying purchasing patterns to make useful recommendations.

Customer Groups

The analysis identified four broad customer segments:

1. Home & Utility
2. Novelty & Party
3. Bags & Storage
4. Decorative Homeware

These groups give the marketing team a simple way to target campaigns around different purchasing patterns.

Risks and Next Steps

The system will be less effective for new customers with little or no purchase history, and recommendations for rare products may be weaker. Customer behavior may also change over time, so recommendation quality should be monitored against the best-seller baseline.

I recommend testing the SVD recommender as a pilot, refreshing recommendations regularly, and tracking Hit Rate@10 to confirm that personalization continues to outperform the simple best-seller approach.